

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (AIML) (Credit System) Degree Examinations**

January 2023

20AM502 – MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) Explain the types of classification techniques in machine learning with an example
- b) Describe data aggregation used in data preprocessing and state the advantages and dis-advantages of it
- c) For the confusion matrix, calculate the Accuracy and Error

8 L*2 1 1

6 L2 1 1

		Predicted Class	
		Class=1	Class=0
Actual Class	Class=1	45 (f_{11})	32 (f_{10})
	Class=0	15 (f_{01})	23 (f_{00})

6 L3 2 2

2. a) Illustrate types sampling approaches
- b) Write the algorithm for KNN and predict the following test data based on KNN

10 L2 1 1

Test Data (T): <Acid Durability=4, Strength=8, Class=?>

Name	Acid Durability	Strength	Class
Type 1	4	3	Good
Type 2	6	5	Bad
Type 3	3	7	Bad
Type 4	2	8	Good

10 L3 2 2

3. a) Write decision tree algorithm and for the dataset, calculate the information gain for the attribute Wind attribute

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

10 L3 2 2

- b) With a neat diagram, explain feature subset selection methodology

10 L2 2 1

Unit – II

4. a) Explain association analysis and apply Apriori algorithm to find frequent 3 itemset for the following data with minimum support of 60%

TID	Bread	Milk	Chocolates	Apple	Eggs	Cola
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

- b) Explain support vector machine algorithm, elaborate on the terminologies: Hyperplanes, support vector, linearity type, non-linearity type, marginal distances
5. a) Write K-means algorithm and solve the following problem
Apply K-Means algorithm to create 2 clusters by assigning each point to the nearest centroid for the given one dimensional point: {6, 12, 18, 24, 30, 42, 48}. The initial centroids can be chosen as {12, 42}.

10 L3 3 2

10 L2 4 1

- b) Explain DBSCAN algorithm with an example
6. a) Describes naïve bayes classification
b) Elaborate on linear regression analysis and multiple regression analysis

10 L3 4 2

10 L2 4 1

10 L2 3 1

10 L2 3 1

Unit – III

7. a) How does random forest works? Write the applications of random forest algorithm in brief
b) Explain in detail ROC curve and its use in machine learning
8. a) Explain dimensionality reduction using factor analysis
b) Describe class imbalance problem

10 L2 5 1

10 L2 5 1

10 L2 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
